MIL-D-24304A(SH)
AMENDMENT 4
8 November 1984
SUPERSEDING
AMENDMENT 3
3 April 1979

MILITARY SPECIFICATION

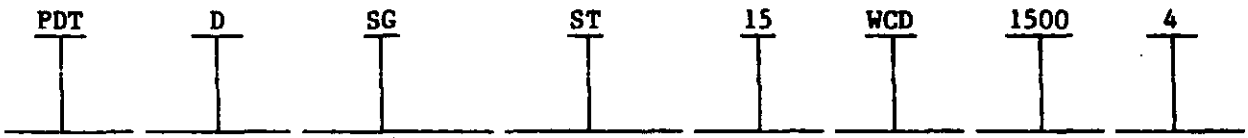
DIFFERENTIAL PRESSURE TRANSDUCER EQUIPMENT
(ELECTRICAL) (NAVAL SHIPBOARD USE)

This amendment forms a part of Military Specification MIL-D-24304A(SHIPS), dated 21 July 1969, and is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department Defense.

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1.2.1: Delete and substitute:

"1.2.1 Type designation. The type designation shall be of the following format and as specified (see 6.3):



Type (see 1.2.1.1)	Sensing technique (see 1.2.1.2)	Transduction principle (see 1.2.1.3)	Application (see 1.2.1.4)	Range (see 1.2.1.5)	Units (see 1.2.1.6)	Pressure rating (see 1.2.1.7	Power supply (see 1.2.1.8)"

* 1.2.1.6 line 3, delete "Psid" and substitute "lb/in^2".

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1.2.1.8: Delete and substitute:

"1.2.1.8 Power supply. Wiring of the power supply shall be one of the following (see 3.5.4):

2 - Two-wire system (direct current (d.c.) supply).
4 - Four-wire system (alternating current (a.c.) supply)."

FSC 6685

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2.1, SPECIFICATIONS, MILITARY, delete: "MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirement For."

2.1, SPECIFICATIONS, MILITARY, add:

"MIL-E-917 - Electric Power Equipment, Basic Requirements For (Naval Shipboard Use).
"MIL-P-24423 - Propulsion and Auxiliary Control Consoles and Associated Control and Instrumentation Equipment, Naval Shipboard Use, Basic Design Requirements."

* 2.1, STANDARDS, MILITARY, add:

"MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
"MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of."

* 2.1, DRAWINGS, add:

"NAVSEA 803-5959187 - Pressure/Differential Pressure Transducer Equipment Application/Selection/Installation Guide."

2.1, HANDBOOK, MILITARY, delete: "MIL-HDBK-H-1 - Cross Index of Chemical Equivalent Specifications and Identification Code (Ferrous and Nonferrous Alloys)."

3.5 and 3.5.1: Delete and substitute:

"3.5 General requirements. Transducer equipment shall be in accordance with the requirements of MIL-P-24423 and as specified herein. When a requirement of MIL-P-24423 conflicts with a requirement of this specification, the requirement of this specification shall govern."

* 3.5, line 4: After "shall govern" add "and shall also be in accordance with Drawing 803-5959187."

"3.5.1 Materials. Materials shall be in accordance with MIL-E-917. Whenever a requirement of MIL-E-917 conflicts with a requirement of this specification the requirement of this specification shall govern."

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3.5.4: Delete and substitute:

"3.5.4 Output. The electrical signal output of the transducer shall be d.c. directly proportional to differential pressure input. The output shall be a true current source of 4 to 20 milliamperes (mA), d.c. The static error band requirement shall be met (see 3.11.1), regardless of external load resistance variations over a span of 0 to 250 ohms."

3.6, item (c), line 2: Delete "5°" and substitute "0°C".

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3.6.7: Add: "The high and low pressure connections shall be clearly labeled by marking on the detector body adjacent to the connections."

3.6.8: Delete and substitute:

"3.6.8 Power supply requirements. Nominal steady state power supply requirements for a.c. shall be 115 volts plus or minus 10 percent, 60 hertz (Hz) plus or minus 5 percent, single phase. Nominal steady state power supply requirements for d.c. shall be 28 volts plus 5 percent minus 3 percent. Transducer shall be capable of operating with power supply variations as specified in 4.8.7 and 4.8.10."

* 3.6.12, line 2: Delete "Naval Ship Engineering Center (NAVSEC)." and substitute "Naval Sea Systems Command (NAVSEA)."

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* 3.7.4.1, line 2: Delete "NAVSEC" and substitute "NAVSEA".

* 3.7.4.2, line 3: Delete "NAVSEC" and substitute "NAVSEA".

* 3.8.2, line 2: Delete "NAVSEC" and substitute "NAVSEA".

* 3.8.3, lines 1 and 5: Delete "NAVSEC" and substitute "NAVSEA".

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* Table I. Material: Delete and substitute:

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"TABLE I. Material.

Service	Nickel-copper	Inconel	Copper-nickel	Alloy steel	Corrosion resisting steel	Ni span	K-monel	Hastelloy C 276
Steam	X	X	X	X	X	X	X	X
Oil	X	X	X	X	X	X	X	X
Seawater	X	X	X	—	—	—	X	X
Flue gas and ammonia	—	—	—	X	X	—	—	X
Oxygen	X	—	X	—	—	—	X	—
Other gases, fresh water, and condensate	X	X	X	—	X	X	X	X

"

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3.11.7: Delete and substitute:

"3.11.7 Supply voltage and frequency or supply voltage (for d.c.) (steady state). Maximum difference between outputs at any voltage and frequency or voltage (for d.c.) condition and the normal (115 volts, 60 Hz or 28 volts d.c.) at the same input, differential pressure and ambient temperature shall not exceed 1.0 percent (see 4.8.7)."

3.11.11, between "frequency" and "(transient)" add: "or supply voltage (for d.c.)".

3.11.12: Delete and substitute:

"3.11.12 Temperature. During the temperature test, transducer performance shall be within the static error band specified in 3.11.1 and 3.11.8 (see 4.8.11)."

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3.11.18: Delete.

* Add new paragraph 3.11.25:

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"3.11.25 Electromagnetic interference emission and susceptibility (EMI). The differential pressure transducers shall meet the following requirements of MIL-STD-461: CE01, CE03, CS01, CS02, CS06, CS09, RE01, RS01, RS02, RS03, except as modified below:

(a) Requirement CS02 shall be met as specified except as follows:

(1) At frequencies between 2 MHz and 30 MHz, the full output of the generator (source impedance of 50 ohms, and an output capability of 12.25 volts rms, or 3 watts, when connected to a 50-ohm load) shall be applied to all power and signal leads of the test sample.
(2) The "calibrating resistor", or 50-ohm load, shown in the CS02 test block diagram of MIL-STD-462, shall not be connected during the test.
(3) The test sample output signal shall not vary more than an amount equal to the static error associated with the accuracy grade (0.5 percent to 5 percent) of the transducer under test.
(4) The test signal shall be applied to all power and signal leads of the test sample."

(b) Requirement RS03 shall be met as specified except as follows:

(1) Between the frequencies of 14 KHz and 18 GHz, the electric field strength test level shall be 10 V/M.

(c) Requirement CS06 shall be met as specified except as follows:

(1) The test signal shall be applied only to AC power leads of the test sample.

(d) Requirements CE01 and CE03 shall be met as specified except as follows:

(1) The test signal shall be applied only to AC power leads of the test sample."

* 4.3, line 2: Delete "NAVSEC" and substitute "NAVSEA".

* 4.3, table III: At the bottom of table III after

" Burst pressure	3.11.22	4.8.22	" add
" Electromagnetic interference emission and susceptibility	3.11.25	4.8.25	"

Tables III and IV: Delete "Dielectric strength".

4.4, lines 1 and 3: Delete "pressure" and substitute "differential pressure".

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Table V: Delete "Dielectric strength".

* 4.5.1, line 2: Delete NAVSEC and substitute "NAVSEA".

4.7.1, items (a), (c) and (d): Delete and substitute:

"(a) Ambient temperature shall be 25 ± 2°C.
"(c) Supply voltage shall be 115 volts a.c. or 28 Vd.c. as required.
"(d) Supply frequency shall be 60 Hz or d.c. as required."

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4.8.7, between "frequency" and "(transient)" add: "or supply voltage (for d.c.)."

4.8.7, line 2: Delete "3.6.8" and substitute "and frequencies (see 3.6.8)".

4.8.9: Delete and substitute:

"4.8.9 Frequency response. The sine wave pressure input signal used shall have a peak to peak magnitude of not less than 1 percent of the input span of the transducer tested. The entire sine wave shall be within the transducer span. Magnitude ratio and phase angle shall be determined over a frequency range from 0.01 Hz to 20 Hz in at least 10 equally-spaced intervals plotted on a logarithmic scale. Performance shall conform to 3.11.10."

4.8.10, line 1, between "frequency" and "(transient)" add: or supply voltage (for d.c.)".

4.8.10.1.2: Delete and substitute:

"4.8.10.1.2 With the transducers operating at the lower limit of steady state a.c. voltage, the a.c. powered transducer shall have a transient voltage of minus 20 percent of nominal voltage recovering to the steady state band in 2 seconds superimposed. With the transducer operating at the lower limit of steady state d.c. voltage and d.c. powered transducer shall have a transient voltage of minus 1 volt recovering to the steady state band in 2 seconds superimposed."

4.8.10.2: Delete and substitute:

"4.8.10.2 Transient frequency (for a.c. powered transducers). A transient frequency of 2 Hz shall be applied of which no more than 1/2 Hz is outside of the steady state band recovering to the steady state band in 2 seconds. This shall be repeated with a minus 2 Hz transient frequency. Performance shall conform to 3.11.11."

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* 4.8.13, line 3: Delete "130 percent of the range for the period of 1 hour" and substitute "200 percent of their range with a maximum pressure of 10,000 lb/in^2 for a period of 1/2 hour."

* 4.8.13, line 4: Delete "psi" and substitute "lb/in^2".

* 4.8.13, line 5: Delete "1 hour" and substitute "1/2 hour".

* 4.8.13, line 7: After "shall conform to 3.11.14." add "When the line pressure rating exceeds 200 percent of the differential pressure range, the overpressure test shall be omitted and the 1 percent deviation allowed in 3.11.14 shall be applied to the line pressure test."

4.8.14, line 2: After "port" insert "with the low pressure port vented to atmosphere".

4.8.18 through 4.8.18.2: Delete.

4.8.19, line 2: Delete "500 volts dc" and substitute 50 volts d.c.".

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* 4.8.22, line 2: Delete "200 percent of the pressure rating, with a maximum of 8,000 psi" and substitute "300 percent of the range with a maximum pressure of 10,000 lb/in^2."

* Add new paragraph 4.8.25:

"4.8.25 EMI tests. EMI tests shall be in accordance with the test methods specified in MIL-STD-462, with the modifications as stated in 3.11.25. Performance shall be as specified in 3.11.25."

* Add new paragraph 4.10:

"4.10 Retention of qualification. To retain qualification, the contractor shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery, indicating as a minimum the number of lots that have passed, the number that have failed, and the group which they failed. The results of tests of all reworked lots shall be identified and accounted for.

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(b) A summary of the results of tests performed for periodic inspection, including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 12-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list."

* Add new paragraph 4.10.1:

"4.10.1 Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 12-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification."

* Add new paragraph 4.10.2:

"4.10.2 In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit his qualified products to testing in accordance with the qualification inspection requirements and the reason for no production."

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* 6.4, line 8: Delete "Naval Ship Engineering Center, Department of the Navy, Center Building, Prince George's Center, Hyattsville, Maryland 20782" and substitute "Naval Sea Systems Command, Department of the Navy, Washington, DC 20362."

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LAST PAGE

* DD Form 1426, Standardization Document Improvement Proposal: Delete address and substitute:

"COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362"

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
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